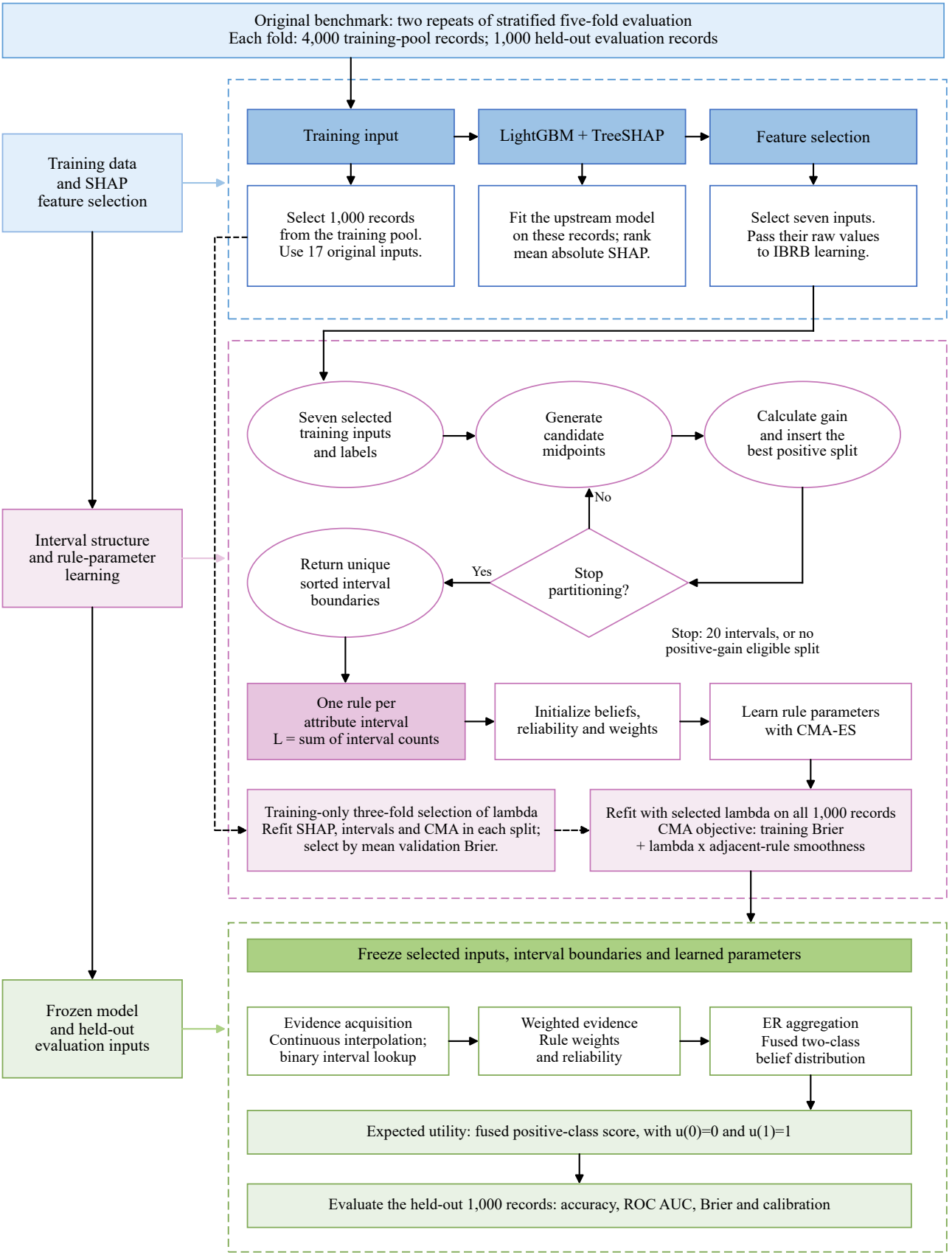
Original benchmark: two repeats of stratified five-fold evaluation
Each fold: 4,000 training-pool records; 1,000 held-out evaluation records
Training data and SHAP feature selection
Training input
LightGBM + TreeSHAP
Feature selection
Select 1,000 records from the training pool. Use 17 original inputs.
Fit the upstream model on these records; rank mean absolute SHAP.
Select seven inputs. Pass their raw values to IBRB learning.
Interval structure and rule-parameter learning
Seven selected training inputs and labels
Generate candidate midpoints
Calculate gain and insert the best positive split
No
Return unique sorted interval boundaries
Yes
Stop partitioning?
Stop: 20 intervals, or no positive-gain eligible split
One rule per attribute interval
L = sum of interval counts
Initialize beliefs, reliability and weights
Learn rule parameters with CMA-ES
Training-only three-fold selection of lambda
Refit SHAP, intervals and CMA in each split; select by mean validation Brier.
Refit with selected lambda on all 1,000 records
CMA objective: training Brier + lambda x adjacent-rule smoothness
Frozen model and held-out evaluation inputs
Freeze selected inputs, interval boundaries and learned parameters
Evidence acquisition
Continuous interpolation; binary interval lookup
Weighted evidence
Rule weights and reliability
ER aggregation
Fused two-class belief distribution
Expected utility: fused positive-class score, with u(0)=0 and u(1)=1
Evaluate the held-out 1,000 records: accuracy, ROC AUC, Brier and calibration